

1. Order Form

- Assumptions:
- 1. Employees help any and all different customers.
 - 2. Each customer has one distinct customerNumber.
 - 3. A customer can order multiple parts in one order form.
 - 4. In each transaction, a customer can be served by only one employee. An employee can serve multiple customers.
 - 5. Each part has one distinct partNumber.
 - 6. A cage code is the identifier of the cages (or shelves) that the inventory is stored in.

1NF:

customerName	<u>customerNumber</u>	customerType	<u>date</u>	<u>time</u>	employee	<u>partNumber</u>	name	type	cageCode	quantityOrdered	unitPrice
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

This table is in 1NF because there is no repeating group and there is a composite key that includes customerNumber, date, time, and partNumber. However, this table is not in 2NF because there are partial dependencies. customerName and customerType depend on customerNumber only. employee depends on customerNumber, date, and time, but doesn't depend on partNumber. name, type, cageCode, and unitPrice depend on partNumber only. These fields depend on part of the composite key instead of the primary key as a whole. To transform the table into 2NF, I separate the above table into 4 tables, customer, employee_service, part, and order.

2NF & 3NF:

customer		
<u>customerNumber</u> PK	customerName	customerType
HG54587	Jeff Peterson	Consumer

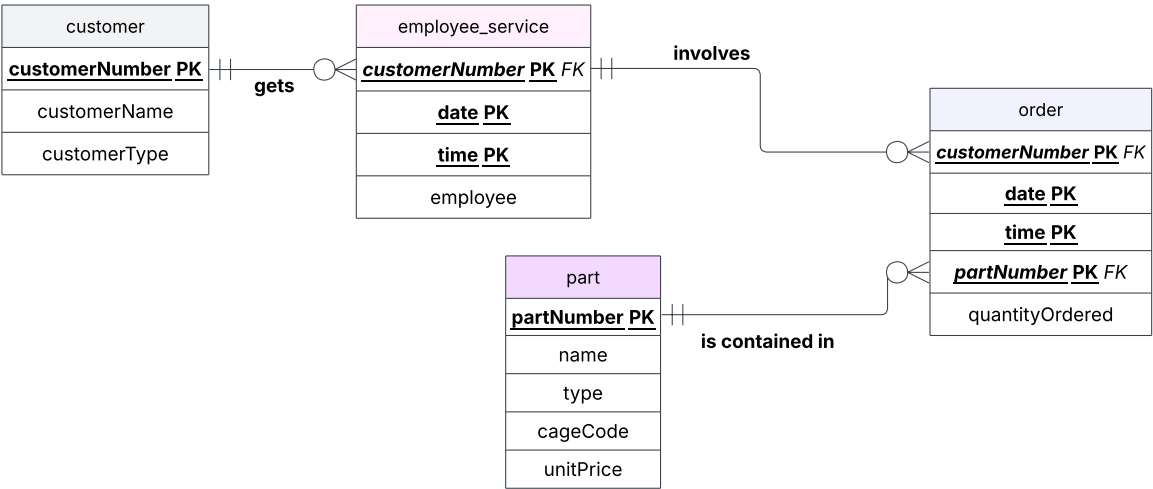
employee_service			
<u>customerNumber</u> PK FK	<u>date</u> PK	<u>time</u> PK	employee
HG54587	7/1/2024	10:30 am	D. Harrison

part				
<u>partNumber</u>	name	type	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order				
<u>customerNumber</u> PK FK	<u>date</u> PK	<u>time</u> PK	<u>partNumber</u> PK FK	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

These tables are in 2NF. They are also in 3NF because there is no transitive dependency in these tables. customerName and customerType depend on customerNumber. employee depends on the composite primary key that includes customerNumber, date, and time. name, type, cageCode, and unitPrice depend on partNumber. quantityOrdered depends on the composite key that includes customerNumber, date, time, and partNumber.

ERD (in 3NF):



2. Therapist Appointments

Assumptions:

1. Each therapist has one unique staffNo and each patient has one unique patNo.
2. A therapist may work at one or more than one branch, but they only see patients at one branch on any appointment day.
3. A therapist sees only one patient at any appointment time.
4. A patient may have multiple appointments in any appointment date and with one or more than one therapist.

1NF:

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

This table is in 1NF because there is no repeating group and there is a composite key that includes staffNo, appointmentDate, and appointmentTime.

However, this table is not in 2NF because there are partial dependencies.

therapistName depends on staffNo only.

branchNo depends on staffNo and appointmentDate, but doesn't depend on appointmentTime.

These fields depend on part of the composite key instead of the primary key as a whole.

To transform the table into 2NF, I separate the above table into 3 tables, therapist, appointment, and therapist_branch_assignment.

2NF:

therapist	
<u>staffNo</u> PK	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

appointment				
<u>staffNo</u> PK	<u>appointmentDate</u> PK	<u>appointmentTime</u> PK	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

therapist_branch_assignment		
<u>staffNo PK</u>	<u>appointmentDate PK</u>	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

These tables are in 2NF, but not in 3NF. There is transitive dependency in the "appointment" table.

In the "appointment" table, patName depends on patNo instead of the compisite primary key that includes staffNo, appointmentDate, and appointmentTime.

To transform the tables into 3NF, I separate the table "appointment" into 2 tables, appointment and patient.

3NF:

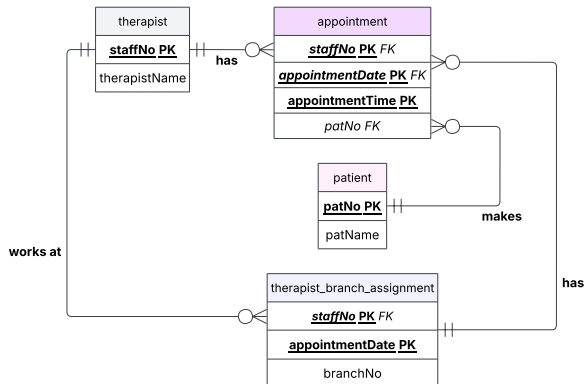
therapist	
<u>staffNo</u> PK	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient	
<u>patNo</u> PK	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

appointment			
<u>staffNo</u> PK <i>FK</i>	<u>appointmentDate</u> PK <i>FK</i>	<u>appointmentTime</u> PK	<i>patNo FK</i>
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

therapist_branch_assignment		
<u>staffNo</u> PK FK	<u>appointmentDate</u> PK	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

ERD (in 3NF):



3. Event Staff

Assumptions:

- 1. Each member of staff has one unique eNo. eName represents the name of each member of staff.
- 2. Each contract applies to one event only.
- 3. Each event has one event location only.
- 4. There may be different contracts for an event depending upon different service needs.

1NF:

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

This table is in 1NF because there is no repeating group and there is a composite key that includes eNo and contractNo. However, this table is not in 2NF because there are partial dependencies. eName depends on eNo only and doesn't depend on contractNo. eventNo and eventLoc depend on contractNo only and don't depend on eNo. These fields depend on part of the composite key instead of the primary key as a whole. To transform the table into 2NF, I separate the above table into 3 tables, staff, contract, and staff_contract_assignment.

2NF:

staff		contract			staff_contract_assignment		
eNo	eName	contractNo	eventNo	eventLoc	eNo	contractNo	hours
1135	Smith J	C1024	H25	Queens	1135	C1024	16
1057	Hocine D	C1025	H4	Yonkers	1057	C1024	24
1068	White T	C1026	H25	Queens	1068	C1025	28
					1135	C1025	15
					1135	C1026	10

These tables are in 2NF, but not in 3NF. There is a transitive dependency in the "contract" table. In the "contract" table, eventLoc depends on eventNo instead of contractNo. To transform the tables into 3NF, I separate the table "contract" into 2 tables, contract and event.

3NF:

staff		contract		event		staff_contract_assignment		
eNo PK	eName	contractNo PK	eventNo FK	eventNo PK	eventLoc	eNo PK FK	contractNo PK FK	hours
1135	Smith J	C1024	H25	H25	Queens	1135	C1024	16
1057	Hocine D	C1025	H4	H4	Yonkers	1057	C1024	24
1068	White T	C1026	H25			1068	C1025	28
						1135	C1025	15
						1135	C1026	10

ERD (in 3NF):

